ICLR 2027 · ORAL

Multi-Fidelity Training Reshapes Uncertainty Decomposition in Atomistic Models

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Code, splits, evaluation
github.com/matteotolloso/mace

Why uncertainty, and why multi-fidelity?

ML potentials are used where nobody checks them, and accurate labels are scarce

Models run beyond their training data

1. **Active learning**
   which configuration do we label next?
2. **Molecular dynamics**
   when does a trajectory leave the model's comfort zone?
3. **Screening**
   which predictions can we keep?

All rely on the model knowing its own error.

Labels come in two fidelities

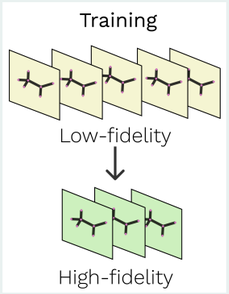
Cheap (DFT)
low fidelity

50,000

Accurate (CCSD(T))
high fidelity

5,000

training configurations per split, ANI-1x



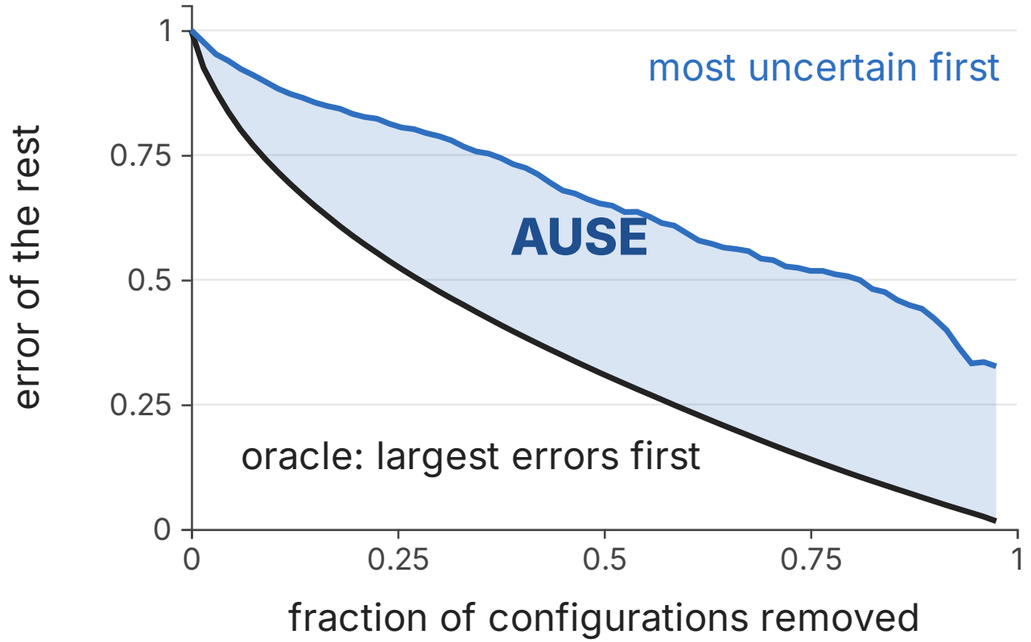
Standard recipe: **pretrain on cheap labels, fine-tune on accurate ones**. It improves accuracy.

Open question: what does multi-fidelity training do to the **uncertainty**, and to its aleatoric/epistemic parts?

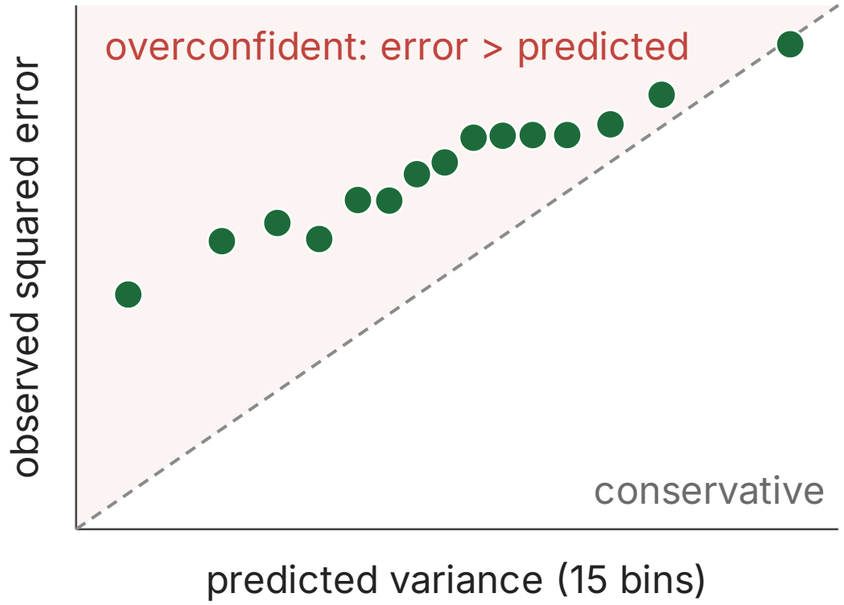
Uncertainty has two jobs

Ranking: which predictions are wrong? · Calibration: by how much?

Ranking: sparsification curve



Calibration: reliability diagram



1 **AUSE = ranking error**

discard the most uncertain configurations and compare with discarding the largest errors; lower is better

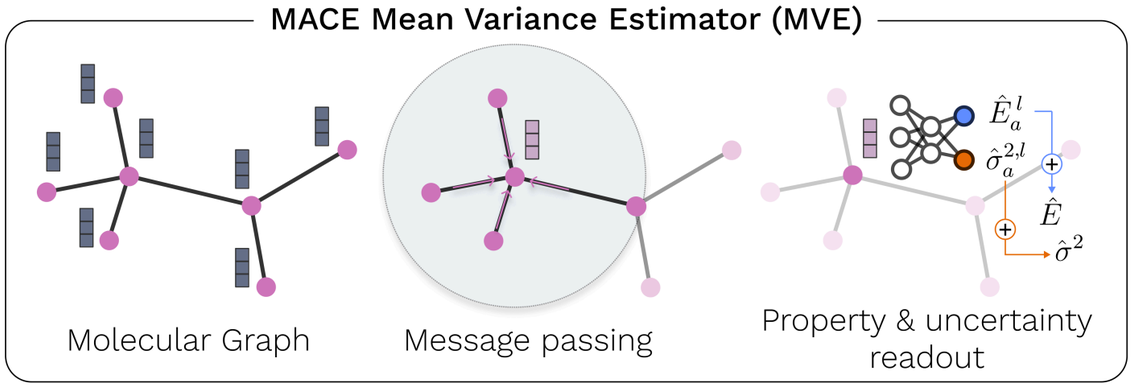
2 **ENCE = calibration error**

predicted variance vs observed squared error in 15 bins; lower is better, above 1 only if overconfident

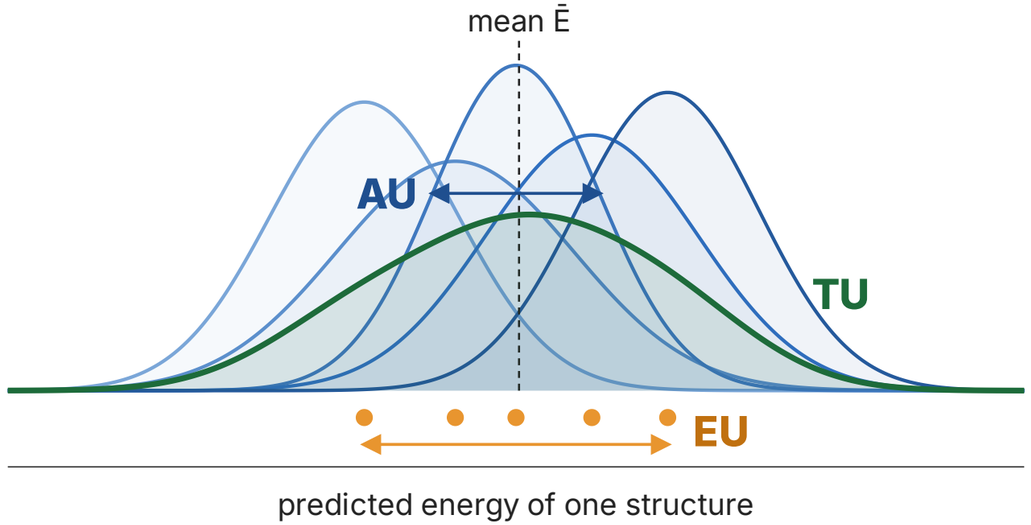
Illustration with synthetic numbers. AUSE: Lind et al. 2024; ENCE: Levi et al. 2022.

MACE models that predict their own error bars

One model gives an energy and a variance; an ensemble of ten splits the uncertainty in two



One model: MACE predicting an energy and its own variance σ², trained with the Gaussian likelihood



Theory: law of total variance

$$\mathrm{Var}(E \mid x) = \mathbb{E}_\theta\big[\mathrm{Var}(E \mid x, \theta)\big] + \mathrm{Var}_\theta\big[\mathbb{E}(E \mid x, \theta)\big]$$

AU · aleatoric

What even a perfect model could not predict: noise or inherent ambiguity. Irreducible.

In practice: average variance of the 10 models

EU · epistemic

Uncertainty about the model itself, from limited data. Reducible with more data.

In practice: disagreement of the 10 models

Practice: ensemble of 10 mean–variance MACE models

$$\mathrm{TU} = \frac{1}{M}\sum_m \hat{\sigma}_m^2(x) + \frac{1}{M}\sum_m \big(\hat{E}_m(x) - \bar{E}(x)\big)^2$$

Labels are exact quantum-chemistry energies: "aleatoric" here means configuration difficulty, not noise. MACE: Batatia et al., NeurIPS 2022.

Experimental setup

Two recipes · two kinds of novelty · uncertainty scored on ranking (AUSE) and calibration (ENCE)

Data

ANI-1x: small organic molecules, each in many shapes

Per split: 50k cheap (DFT) + 5k accurate (CCSD(T)) energy labels

Energies only, no forces

Two recipes

HF-only: accurate labels only, from scratch

LF→HF: pretrain on cheap labels, then fine-tune on accurate ones

Same architecture, 10-model ensembles, 5 data splits

Test sets

In domain (ID): new shapes of training molecules

System shift: molecules never seen

Energy shift: trained on low-energy shapes, tested on high-energy

RQ1 At the end of training, is the uncertainty better?

RQ2 Does the kind of novelty matter?

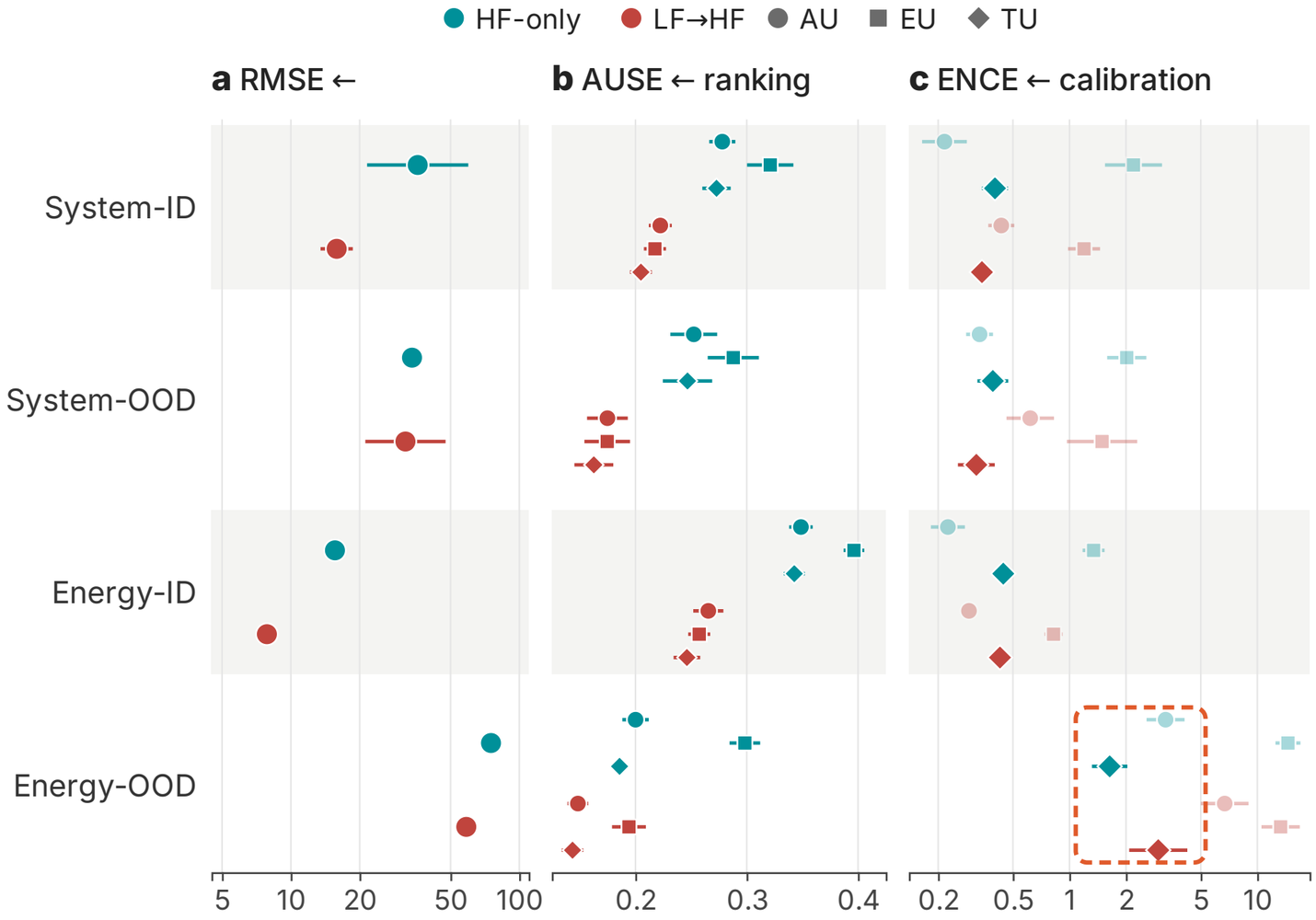
RQ3 How does uncertainty change during training?

RQ4 Is it a better guide for choosing new data?

Pretraining improves ranking, not always calibration

RQ1 · At the end of training, does pretraining give better uncertainty?

1 2 3 4



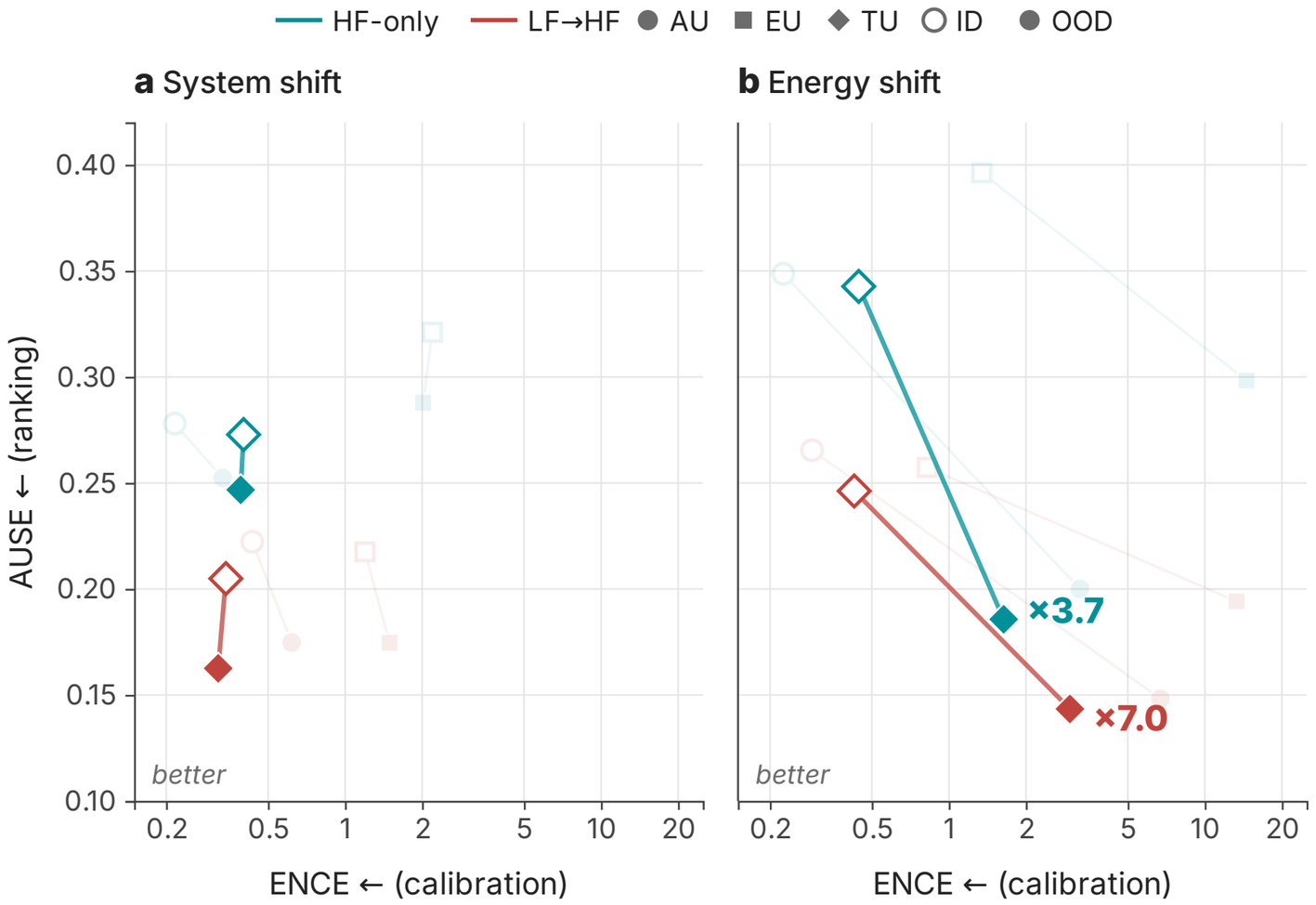
1. **Lower error everywhere**
   more accurate on every test set (a)
2. **Better ranking on every test set**
   for AU, EU and TU alike (b)
3. **EU catches up**
   from weakest signal to close to TU (b)
4. **Calibration is mixed**
   TU worse at high energy (c)

So: pretraining helps the model know **which** predictions are wrong, not always **by how much**.

Teal: HF-only · **Red**: LF→HF · ● AU ■ EU ◆ TU · left is better (RMSE = error, AUSE = ranking error, ENCE = calibration error) · bars: 95% intervals over 5 splits

Energy shift decouples ranking from calibration

RQ2 · Does the kind of novelty matter: new molecules vs higher energies?



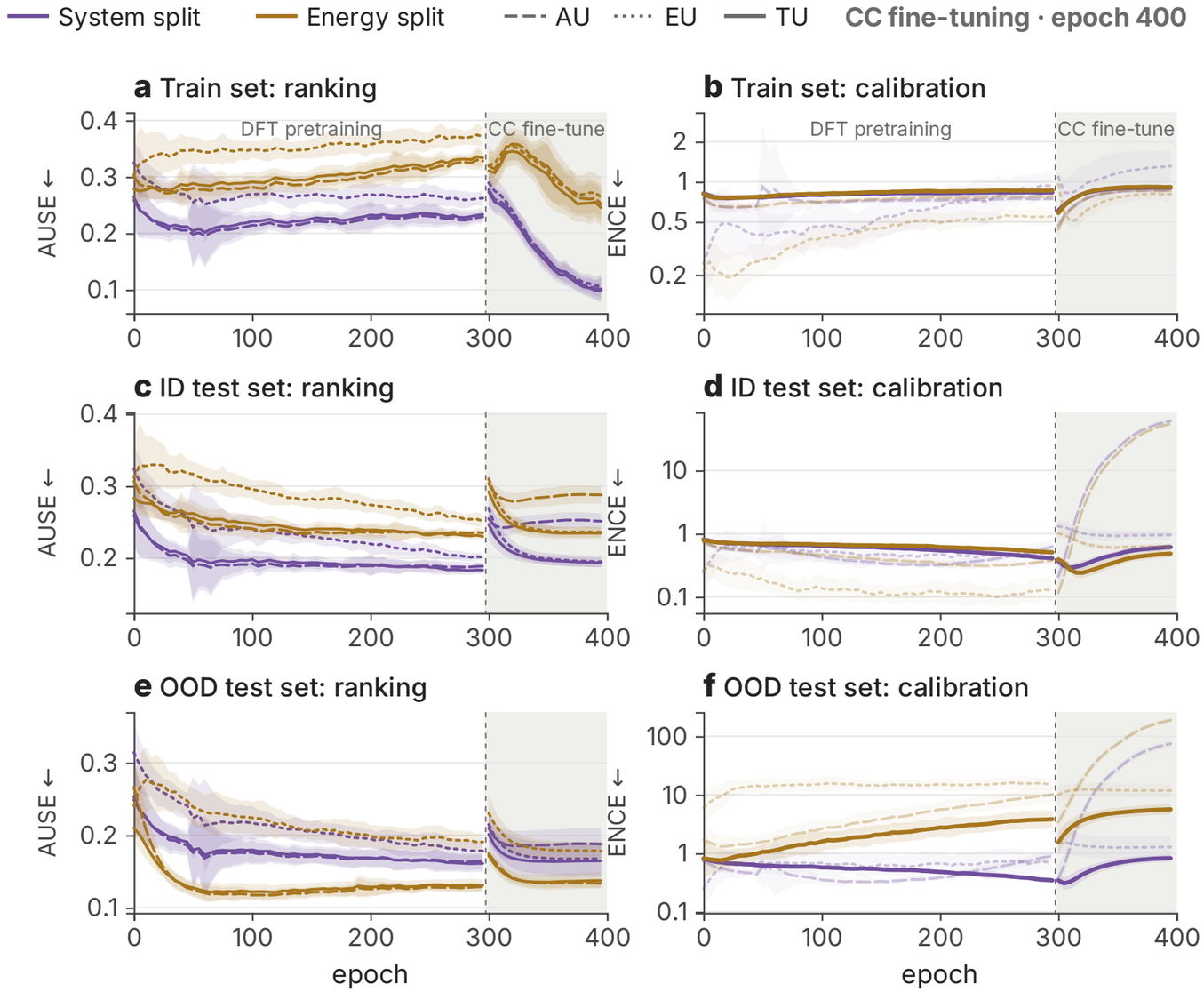
1. **New molecules: ranking improves**
   and calibration barely changes (a)
2. **Higher energies: ranking improves too**
   larger errors are easier to order (b)
3. **But calibration breaks**
   TU calibration error ~4–7× worse (b)
4. **Recalibration does not fix it**
   a fit in domain cannot stretch to bigger errors (backup)

So: the models know **which** high-energy configurations are wrong, but not **by how much**.

Hollow = in domain, filled = out of domain · down = better ranking, left = better calibration · numbers: TU ENCE OOD / ID

Calibration is lost during high-fidelity training

RQ3 · During training, when is calibration lost?



1. **On training data, all is fine**
   ranking improves, calibration stays put (a, b)
2. **On test data, ranking stays good**
   also after switching to accurate labels (c, e)
3. **But calibration breaks in fine-tuning**
   good during pretraining, then much worse (d, f)
4. **The culprit is AU**
   error bars far too small (~100× worse)
5. **EU holds**
   stays near its pretraining level and limits the damage

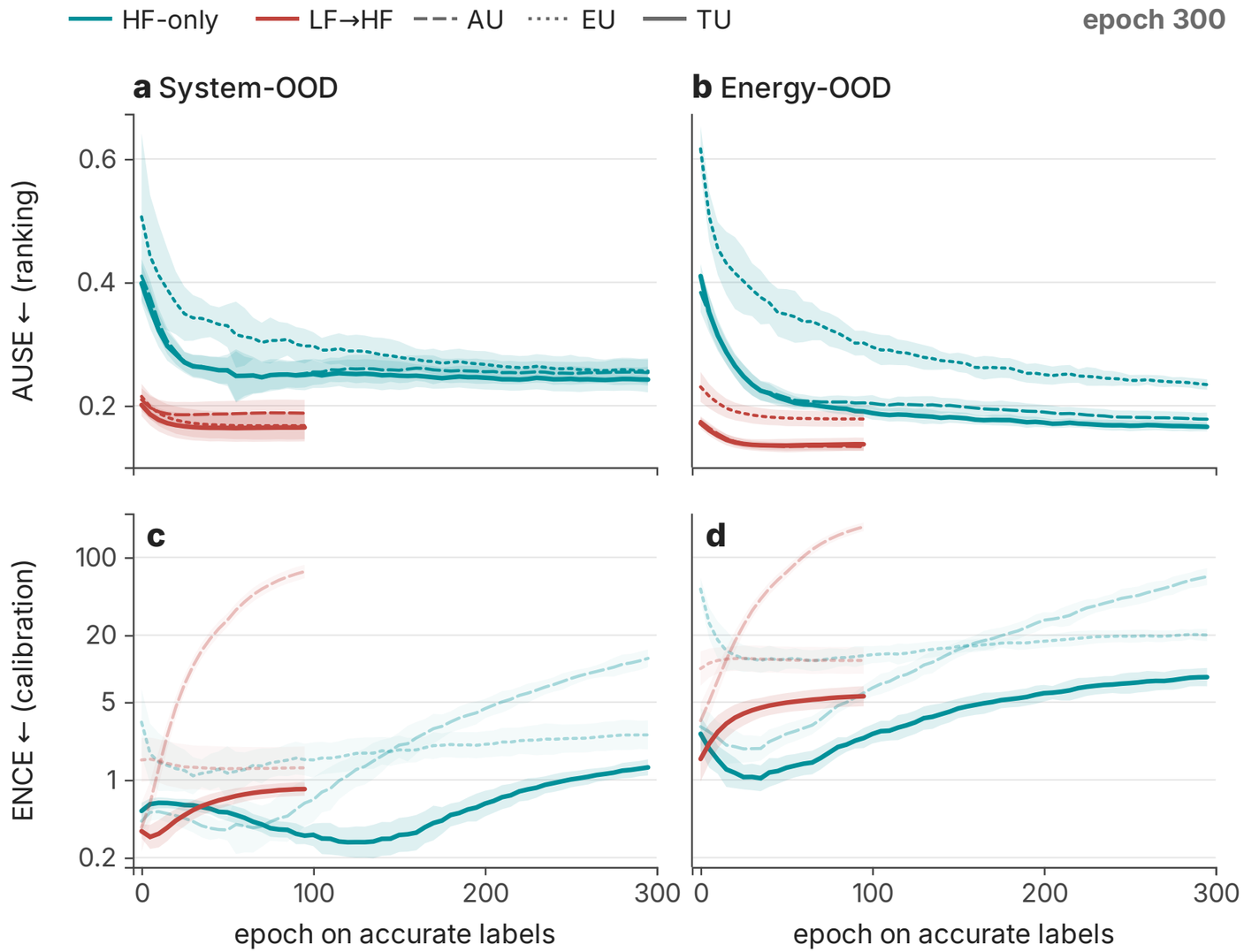
So: most calibration is lost in the accurate-label stage, through AU.

Epochs 0–299: pretraining on DFT labels · shaded 300–399: fine-tuning on CC labels · **purple**: system split, **brown**: energy split · bands: 95% over 5 splits

Pretraining speeds up both gain and loss

RQ3 · Same question, comparing the two recipes on the accurate-label stage

1 2 3 4



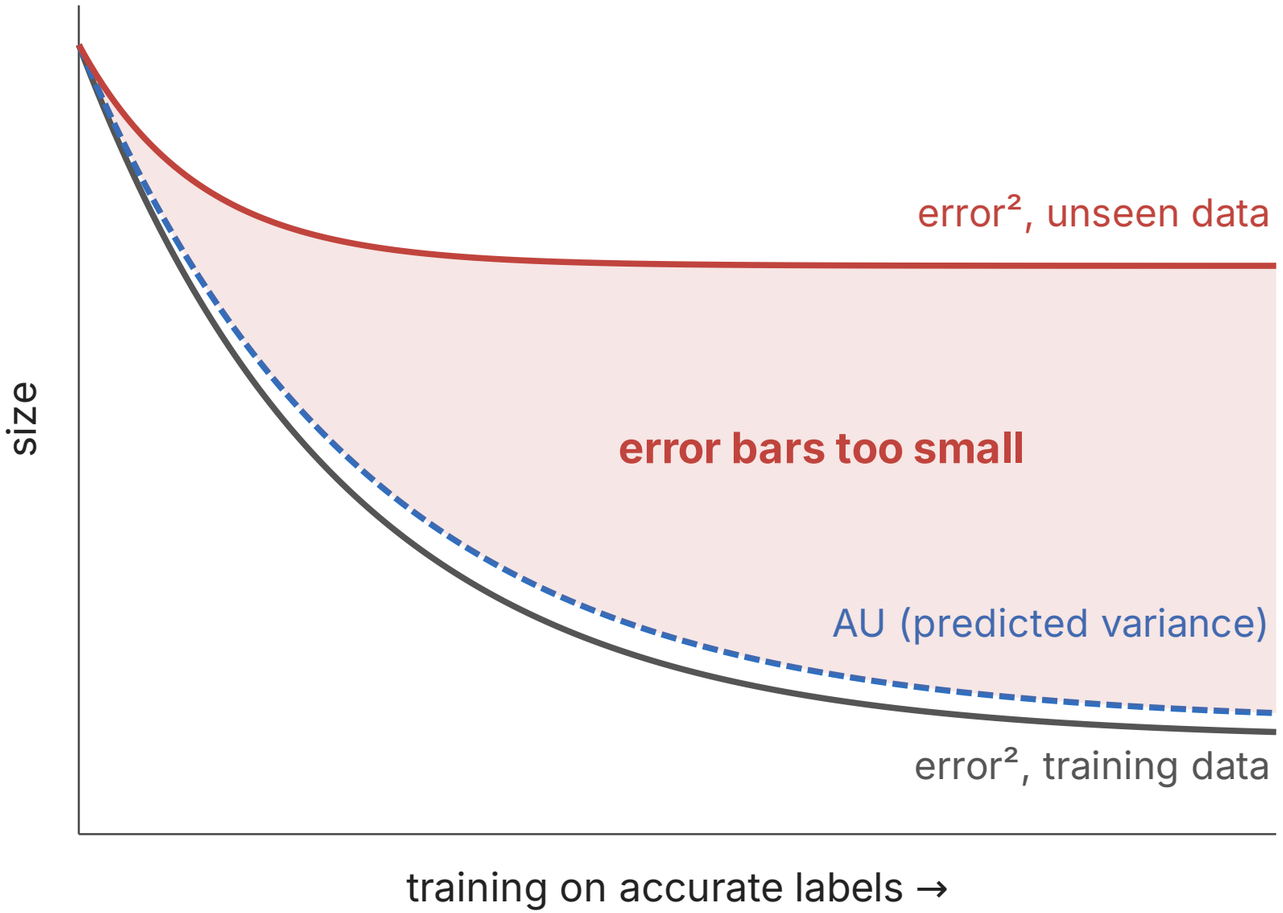
1. **Pretraining ranks better at once**
   from the first accurate-label epoch (a, b)
2. **Then both recipes level off**
   ranking settles during training
3. **Calibration breaks faster**
   AU ENCE ≈80 / 200 after 100 epochs, vs ≈14 / 70 after 300 from scratch (c, d)
4. **Ranking and calibration decouple**
   one can improve while the other gets worse

In practice: select checkpoints by watching calibration too, not only the loss, ideally on shifted data.

x axis: epochs on accurate (CC) labels only · **teal** starts from scratch, **red** from the pretrained model · OOD test sets

Why do the error bars shrink?

A plausible mechanism (our interpretation)



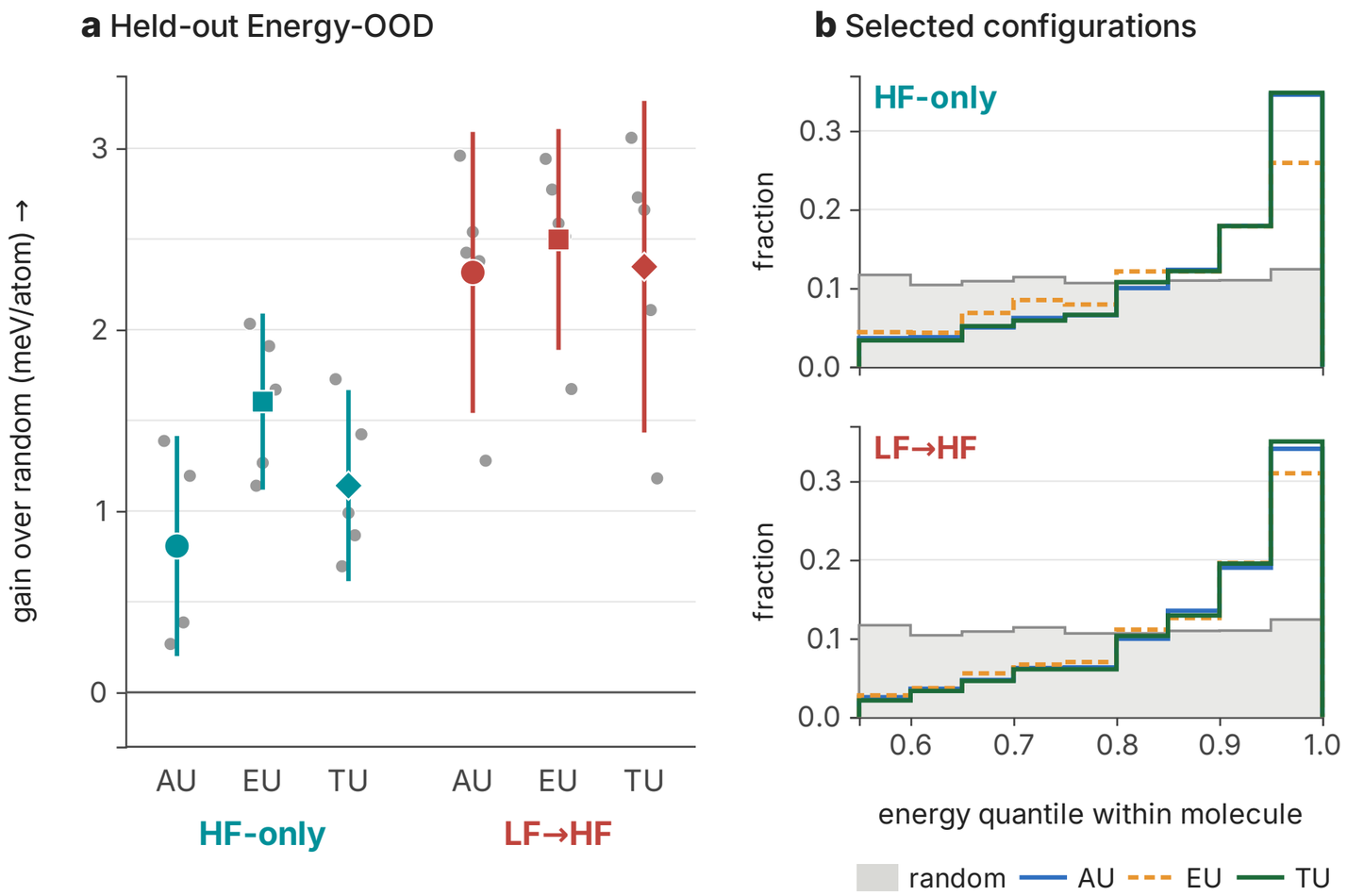
1. **AU learns from training errors**
   each variance head is fitted to its own squared training residuals
2. **Those errors shrink, unseen ones less**
   scarce accurate data are fitted closely
3. **Small error bars carry over**
   overconfident on new data; a known pitfall of mean–variance training

So: with exact labels, AU measures in-sample difficulty, not irreducible noise.

Schematic, not data. Nix & Weigend 1994; Seitzer et al., ICLR 2022.

Pretrained uncertainty is a better acquisition function

RQ4 · Is uncertainty a better guide for choosing which data to label?



1. **Uncertainty beats random**
   for both recipes (a)
2. **Pretraining gives a larger gain**
   2.3–2.5 vs 0.8–1.6 meV/atom (a)
3. **…from a better start**
   already more accurate before acquisition (≈59 vs ≈75 meV/atom)
4. **Same kind of data picked**
   mostly high-energy configurations (b)

So: pretrained uncertainty is the better guide to which simulation to run next.

One round: add 500 of 2,500 high-energy configurations, chosen by uncertainty or at random, retrain, test on the held-out half · a: error drop over random (higher is better), grey dots = splits

Take-home messages

Uncertainty in multi-fidelity models is not a static property

1. **Pretraining improves accuracy and ranking**
   for every signal, and EU becomes a strong ranker

2. **…but not necessarily calibration**
   energy extrapolation: known which, not by how much

3. **Calibration is lost in accurate-label training**
   faster after pretraining; AU turns overconfident, EU mitigates

4. **Better acquisition after pretraining**
   larger gain from uncertainty-guided data selection

In practice

- Score ranking and calibration separately
- Monitor held-out calibration during training; use it to select checkpoints
- Be careful with AU after training on scarce accurate data
- Validate in-domain recalibration before using it under extrapolation

THANK YOU

Questions?

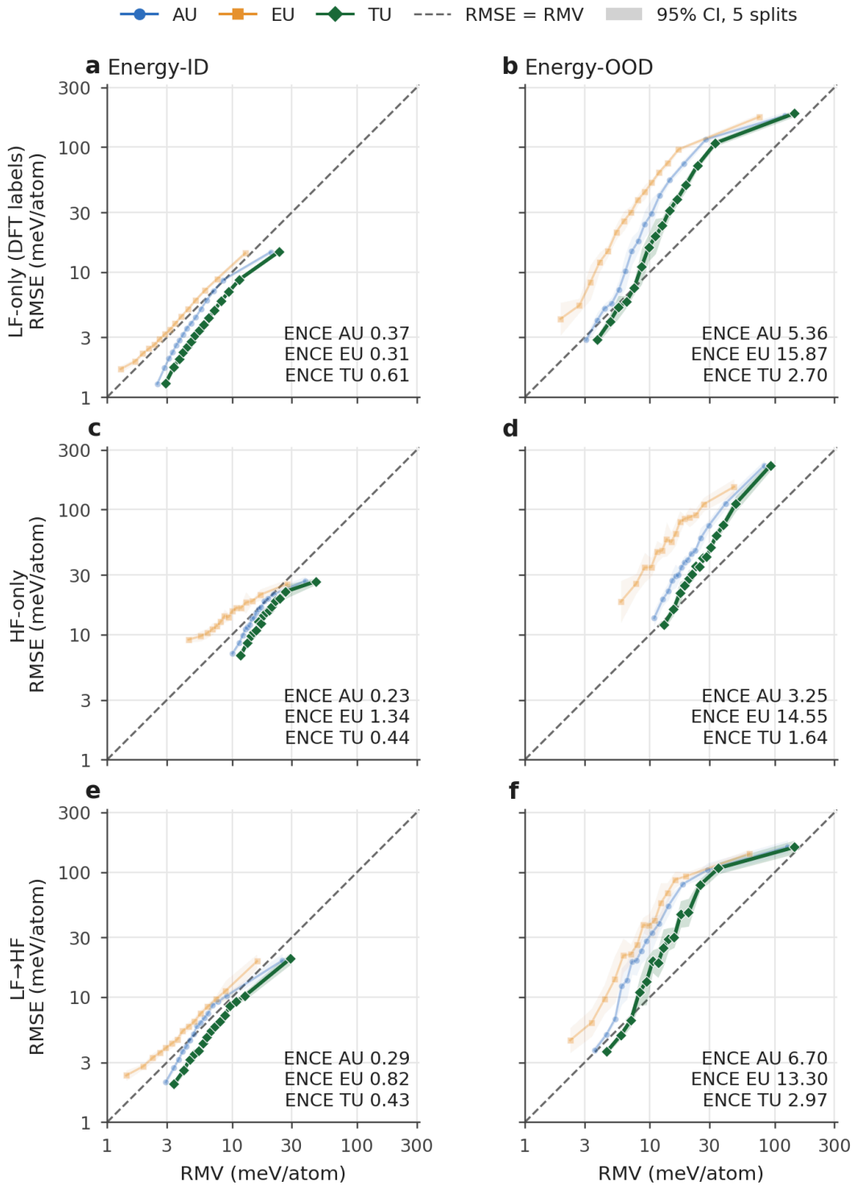
Code & splits: github.com/matteotolloso/mace

Contact: matteo.tolloso@phd.unipi.it

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Backup

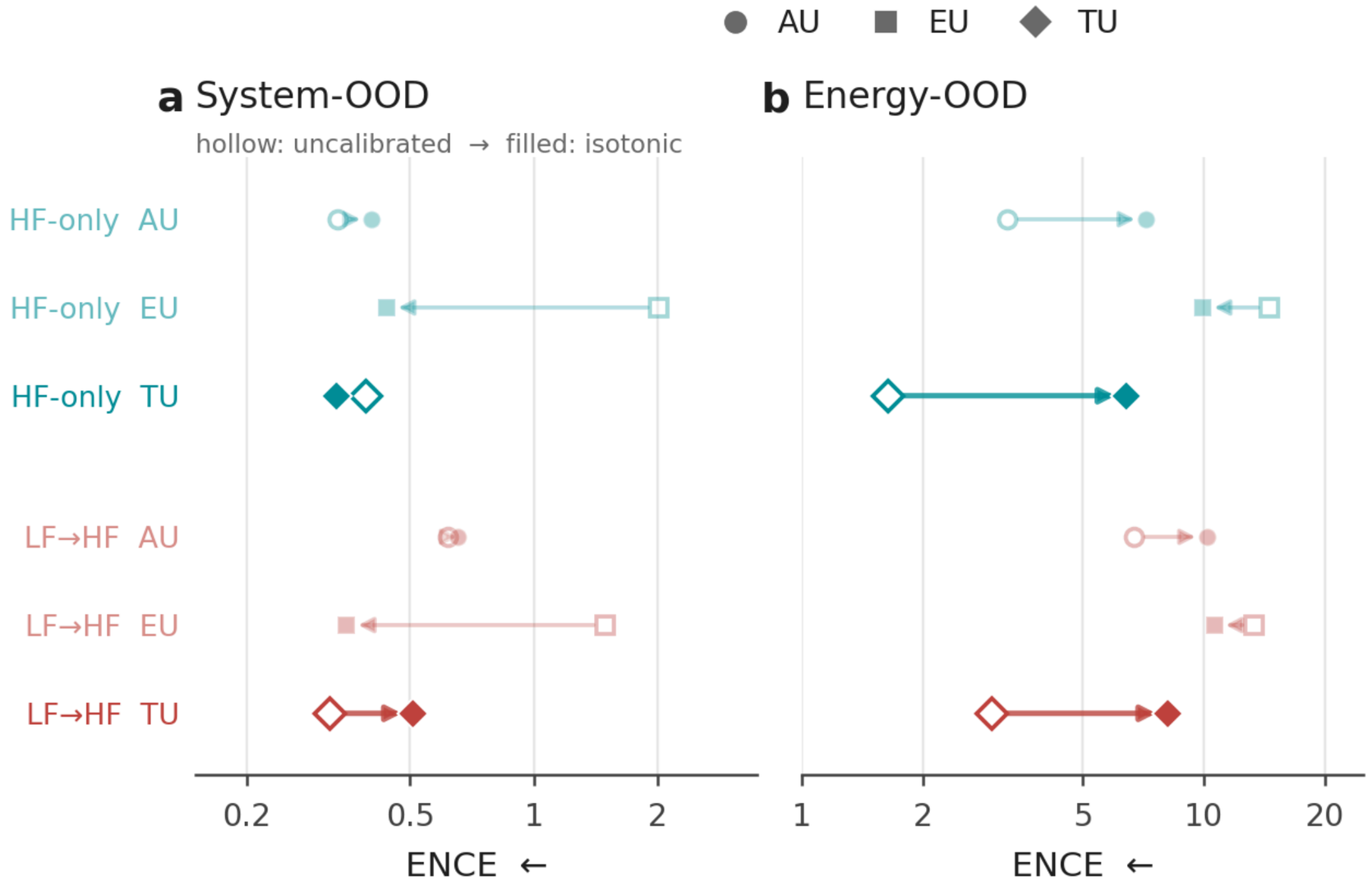
Reliability diagrams, energy split



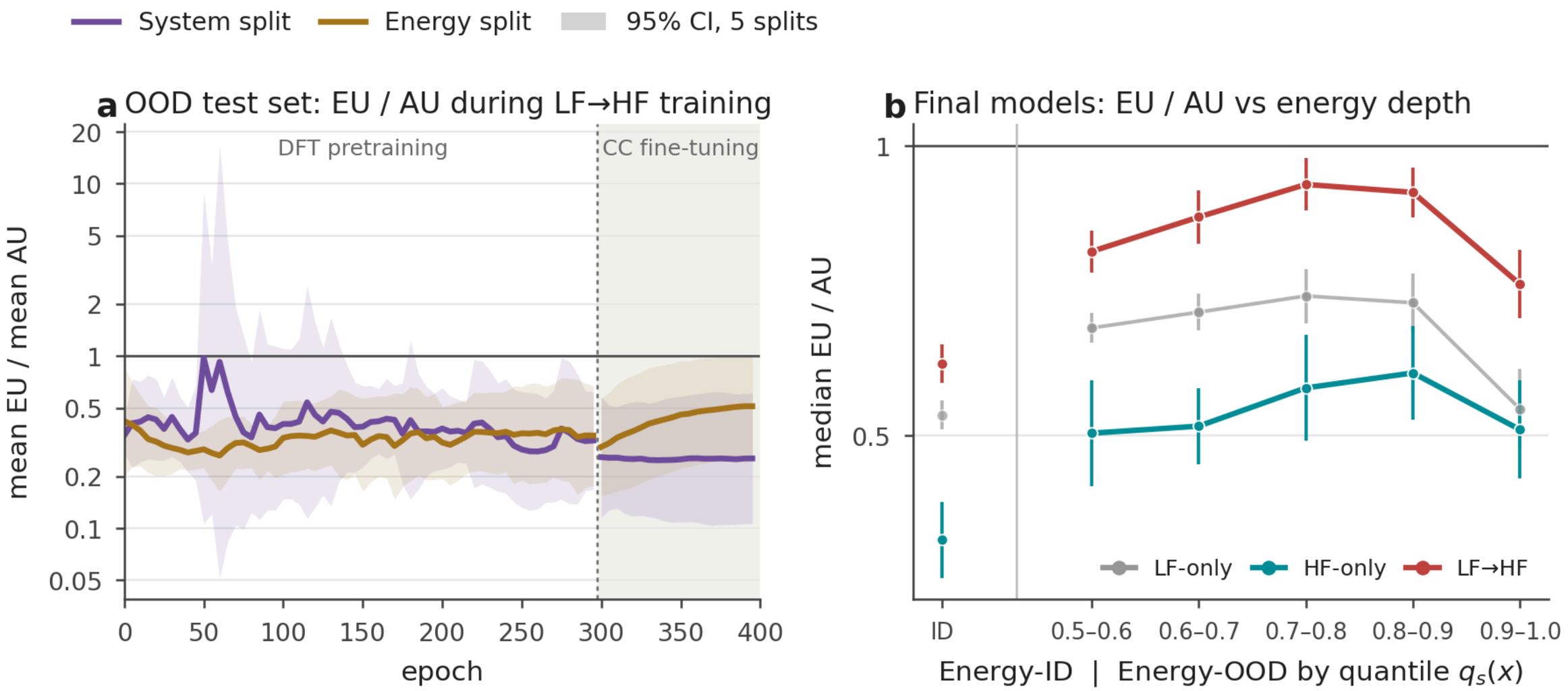
Rows: LF-only (DFT labels), HF-only, LF→HF. Columns: Energy-ID, Energy-OOD. 15 equal-count bins; corner values: geometric-mean ENCE over five splits.

In domain TU lies on or slightly below the diagonal (conservative); on Energy-OOD most bins lie above it (overconfident), EU most of all.

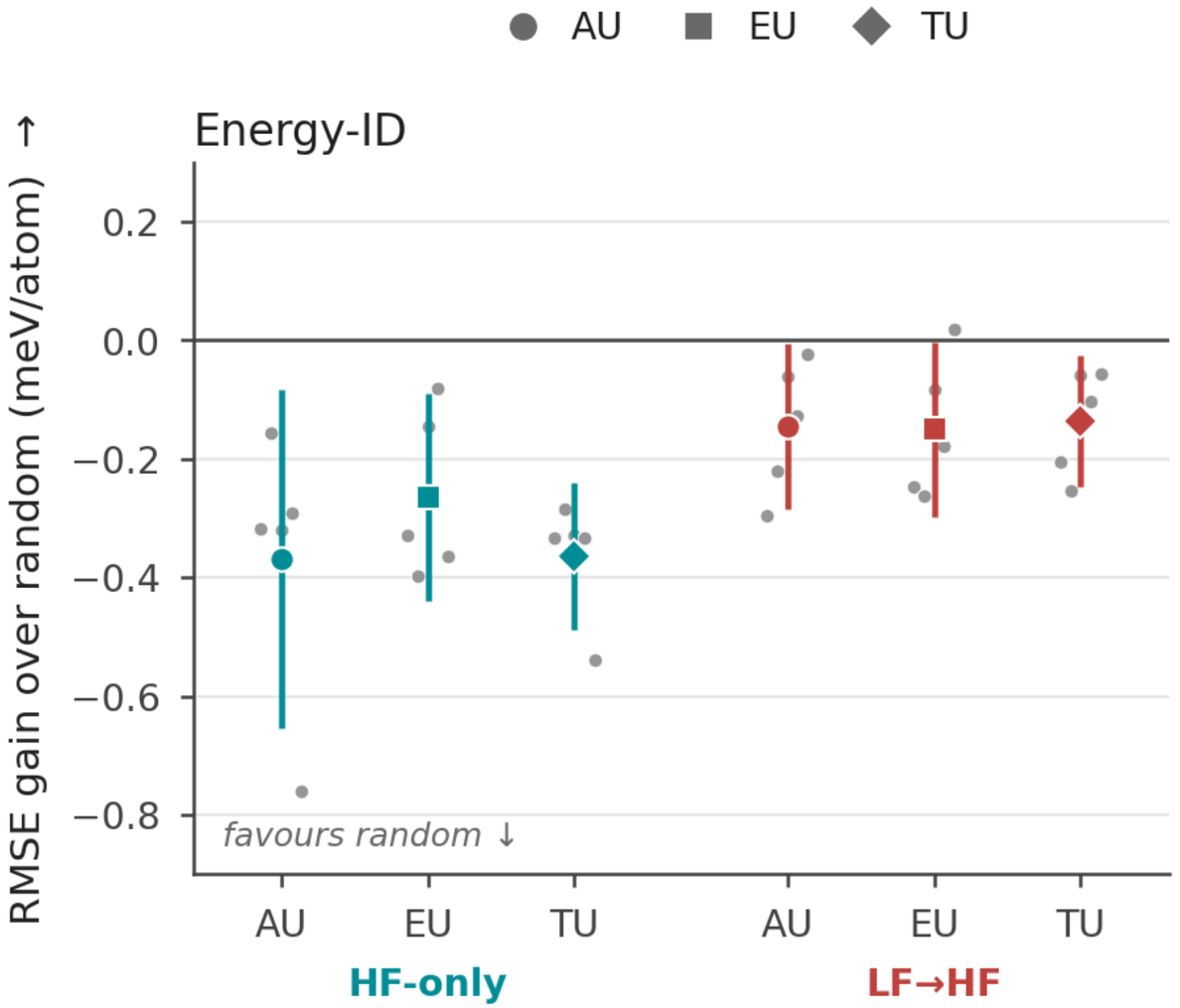
In-domain recalibration does not repair Energy-OOD



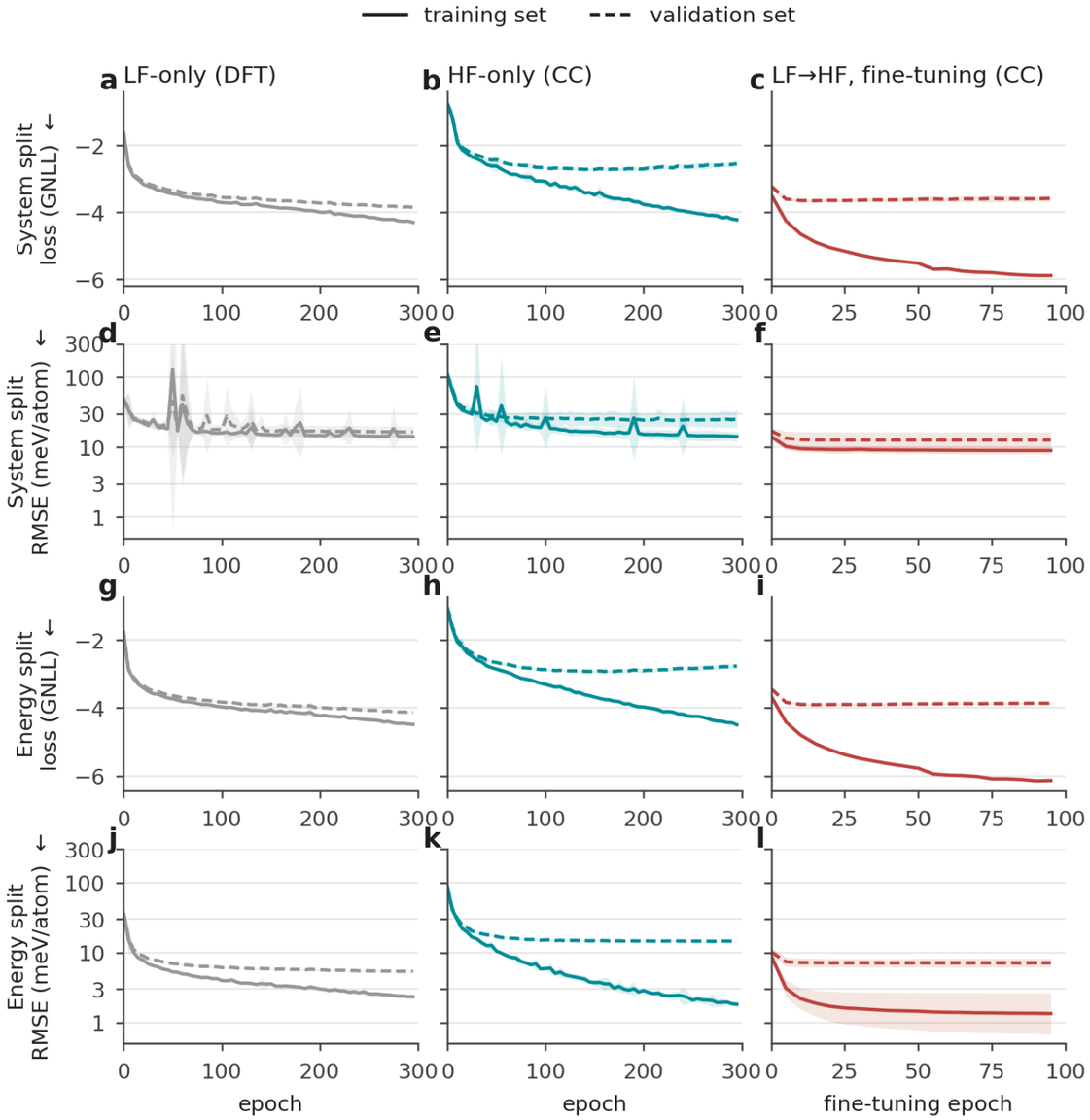
Balance between the components: EU/AU



Acquisition in domain: no gain over random



Training and validation curves



Scope of the study

1. **Energies only**
   force uncertainty is local, vector-valued and drives MD stability

2. **One estimator**
   mean–variance deep ensembles; cheaper alternatives untested; no ablation of the variance head

3. **Operational decomposition**
   no ground truth for AU and EU separately; calibration is judged on TU

4. **Data access**
   LF→HF also sees the DFT labels; not separated from the change of fidelity

5. **One dataset, controlled shifts; one acquisition round**
   real workflows shift composition, size, temperature and energy at once